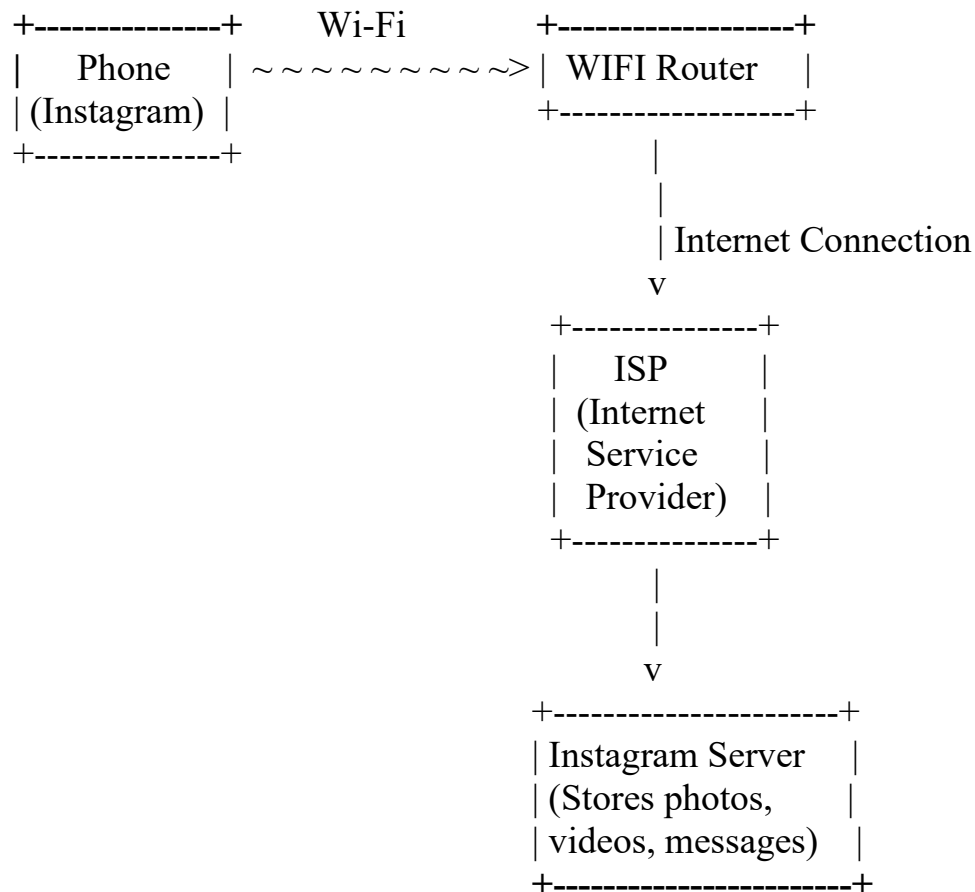


N+ - Network Fundamentals and building network

❖ Introduction to Networks

[Q-1] Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.

[ANS]



➤ Connection flow:

Phone → WiFi Router → ISP → Instagram Server

- Phone: Runs the Instagram app.
- WiFi Router: Connects your phone to the internet.
- ISP: Provides internet access and routes data.

- Instagram Server: Receives requests and sends Instagram content back to your phone.

➤ Data travels in both directions:

- text
- Phone ⇌ WiFi Router ⇌ ISP ⇌ Instagram Server

[Q-2] List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

[ANS]

Device	Network Type	Explanation
Smartphone	WAN (Mobile Network) and LAN (Wi-Fi)	Connects to the internet through cellular networks or a home Wi-Fi network.
Laptop	LAN	Connects to a local network through Wi-Fi or Ethernet at home, school, or work.
Smart TV	LAN	Uses a home Wi-Fi or wired network to stream content from the internet.
Smartwatch	PAN (Personal Area Network)	Often connects to a smartphone using Bluetooth, which is a PAN.
WiFi Router	LAN and WAN	Creates the local network (LAN) and connects it to the internet through the WAN.

[Q-3] Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

[ANS.]

Internet	Intranet
A global network that connects millions of devices worldwide.	A private network used within an organization, such as a college or company.
Accessible to anyone with an internet connection.	Accessible only to authorized users within the organization.
Used for websites, email, social media, and online services.	Used for sharing internal information, notices, files, and resources.

➤ **Example from a College**

- **Internet:** When students use the college Wi-Fi to browse Instagram, watch YouTube, or search Google, they are accessing the Internet.
- **Intranet:** If the college has a student portal where students can check attendance, download course materials, view exam results, or access internal notices, that portal is part of the Intranet because it is intended only for students and staff of the college.

**[Q -4] Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

Hint: Mention the devices and networks involved, and what happens when you hit 'search'..**

[ANS]

➤ **WhatsApp Example:**

- When I search for a chat in WhatsApp and press ****Search****, my phone sends the request through Wi-Fi or mobile data to WhatsApp's servers.
- The servers process the request and find the matching data.
- The results are sent back through the internet to my phone.
- WhatsApp then shows the search results on my screen.